

Weighted and Covered Supports for Mersenne Subseries

The series $\sum_{a \in A} (b^a - 1)^{-1}$ and Erdős Problem #257

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ABSTRACT

We give weighted and divisor-cover conditions ensuring that every infinite Mersenne subseries on a support is irrational at every integer base $b \geq 2$. Both conditions allow divergent reciprocal sums. Averaging over multiples and then over dyadic lengths places both estimates on a common finite distribution, so the conditions can be combined. The resulting class contains supports in neither individual class. Irrationality for arbitrary infinite support in base two remains open.

1 Introduction

For a set A of positive integers and a real number $b > 1$, write

$$X_A(b) = \sum_{a \in A} \frac{1}{b^a - 1}.$$

This series converges by comparison with a geometric series, since $(b^a - 1)^{-1} \leq b^{1-a} / (b - 1)$. Erdős proved irrationality for full support at integer bases [5], and later proved the same conclusion for pairwise coprime supports of finite reciprocal sum [7, p. 222]. On that page he stated that the coprimality assumption could be removed, without giving the details. The question for arbitrary infinite support in base two is Problem #257:

Problem 1.1 (Erdős #257). Is $X_A(2)$ irrational for every infinite $A \subseteq \mathbb{N}_{>0}$?

We seek conditions that imply irrationality for every infinite subset of A , without having to analyse the selected subset anew. The first uses the powers of finitely many primes dividing an exponent: a large prime part gives a small weight even when the reciprocal sum of A diverges. If P is a finite set of primes, the P -part of a is $h(a) = \prod_{p \in P} p^{v_p(a)}$, where $v_p(a)$ is the exponent of p in a . For $P = \{2\}$ and $a = 2^k m$ with m odd, the weight below is $1/[m(b^{2^k} - 1)]$.

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Theorem 1.2 (a weighted condition on the support). *Let $b \geq 2$ be an integer, let $A \subseteq \mathbb{N}_{>0}$ be infinite, and let P be a finite nonempty set of primes. Set $h(a) = \prod_{p \in P} p^{v_p(a)}$. If*

$$W_{b,P}(A) := \sum_{a \in A} \frac{h(a)}{a(b^{h(a)} - 1)} < \infty, \quad (1)$$

then $X_A(b)$ is irrational. In particular, the base-two condition

$$\sum_{a \in A} \frac{h(a)}{a(2^{h(a)} - 1)} < \infty \quad (\text{W})$$

implies that $X_A(b)$ is irrational for every integer $b \geq 2$. Both conclusions are hereditary under passage to infinite subsets.

Consider the support

$$A_* = \{2^k m : k \geq 1, m \text{ odd}, m \leq 2^{2^k}\}.$$

Each 2-adic layer has reciprocal mass at least $1/4$, while its contribution to (W) is at most 2^{1-k} . Thus the reciprocal sum diverges and the weighted sum converges. The calculation in Section 2 proves both bounds. Every infinite subset of A_* consequently gives an irrational value at every integer base. Since $h/(2^h - 1) \leq 1$, the same theorem also includes reciprocal-summable supports. Section A gives a direct proof of Erdős's stated extension.

The divisor-cover condition instead bounds fractional powers of finite divisor counts by positive sums over divisors (Theorem 3.1). We prove that the two conditions are incomparable and that their union gives a further irrationality criterion: if E satisfies the weighted condition and V has a summable divisor cover, every infinite subset of $E \cup V$ has irrational sum (Theorem 3.2). The separating constructions in Proposition 4.1 then give a union outside both individual classes (Corollary 4.2). These are sufficient hypotheses for Problem 1.1; the arbitrary support in that problem need not satisfy them.

For the proofs, we compare a multiple of the series with an integer. Define

$$\Delta_{b,A}(N) = \sum_{a \in A} \frac{b^{N \bmod a} - 1}{b^a - 1} = (b^N - 1)X_A(b) - J_{b,A}(N), \quad J_{b,A}(N) = \sum_{\substack{a \in A \\ a \leq N}} \sum_{j=1}^{\lfloor N/a \rfloor} b^{N-j a}. \quad (\text{D})$$

The identity follows by division of N by a . Since $J_{b,A}(N)$ is an integer at an integer base, rationality $X_A(b) = p/q$ would force every positive displacement to be at least $1/q$. Infinitude of A ensures positivity for $N > 0$: there is a nonzero summand with $a > N$. We shall therefore make these displacements arbitrarily small. Erdős also proposed using fractional parts of powers times the series value [7, p. 226].

We first choose a modulus divisible by a finite part of the support; those exponents contribute zero to (D). Averaging over multiples controls the remaining complete residue periods. Their incomplete periods are the difficulty, because the modulus must also grow. A second average over dyadic observation lengths bounds these errors. Its length can be chosen large compared with the modulus and small compared with the

exponential saving supplied by the gcd. The same finite distribution then controls the weighted and covered parts of a union. Finding a small displacement for each part at unrelated indices would leave the mixed assertion unproved.

Section 2 proves the weighted estimate. Section 3 develops the divisor-cover estimate on the same windows and proves the mixed theorem. We construct the separating supports in Section 4 and discuss the limits of the argument in Section 5. The appendices treat the inherited reciprocal-summable theorem, finite denominators and tests for rational membership. Neither $1/2$ nor $1/21$ is decided by those tests.

We shall also use the divisor counts

$$c_A(n) = \#\{a \in A : a \mid n\}, \quad X_A(b) = \sum_{n \geq 1} c_A(n) b^{-n}. \quad (2)$$

Expanding $(b^a - 1)^{-1} = \sum_{j \geq 1} b^{-aj}$ and interchanging nonnegative sums gives the identity. Although $\mathbf{1}_A$ takes only the values zero and one, its divisor transform may take larger values, with $0 \leq c_A(n) \leq \tau(n)$.

2 Finite averages and the weighted criterion

Fix the integer base b and the finite prime set P . We shall choose Q so that the residues of Qm modulo most exponents have a small mean. The gcd controls that mean. For example, when $b = 2$, $Q = 4$ and $a = 6$, the residue orbit is $4, 2, 0$, and

$$\frac{1}{3} \left(\frac{2^4 - 1}{63} + \frac{2^2 - 1}{63} + 0 \right) = \frac{2}{21} \leq \frac{2}{6(2^2 - 1)} = \frac{1}{9}.$$

The final expression is the mean of $2^{4m \bmod 6}/63$ over the same orbit. If $6 \mid Q$, the displacement is zero throughout the orbit. We use these two effects separately: divisibility removes a finite part of A , and a large gcd controls the rest.

Proof. Put $d_a(m) = (b^{m \bmod a} - 1)/(b^a - 1)$ and $g = (Q, a)$. For positive integers Q, a, T , the orbit has length a/g . Splitting $1, \dots, T$ into full orbits and a remainder gives

$$\frac{1}{T} \sum_{t=1}^T d_a(tQ) \leq \frac{g}{a(b^g - 1)} + \frac{1}{T(b^g - 1)}. \quad (3)$$

Indeed, the orbit has length a/g and $\sum_{t=1}^{a/g} b^{tQ \bmod a} / (b^a - 1) = 1/(b^g - 1)$; an incomplete orbit costs at most one more complete orbit. Also, for $Y = QT$,

$$\frac{1}{T} \sum_{t=1}^T \sum_{\substack{a \in A \\ a > Y}} d_a(tQ) \leq \frac{4}{T}, \quad (4)$$

because $d_a(tQ) \leq 2 \cdot 2^{tQ-a}$ when $a > QT$ and $\sum_{t=1}^T 2^{tQ-QT} \leq 2$.

To sum the incomplete-orbit errors while Q grows, we average over dyadic lengths before choosing the observation. If $Q, M \geq 1$ are integers and $\alpha_a \geq 0$ with $\sum_a \alpha_a/a < \infty$, then

$$\sum_{j=M}^{2M-1} \frac{1}{2^j} \sum_{a \leq Q2^j} \alpha_a \leq 2Q \sum_{a \geq 1} \frac{\alpha_a}{a}; \quad (5)$$

for each fixed a , the admissible geometric tail is at most $2Q/a$.

Fix $\varepsilon > 0$. Choose finite nonempty $F \subseteq A$ so that the (1) mass outside F is below ε , and let L be any fixed positive common multiple of F . With $p_* = \max P$ and large $H \geq 2p_*$, set

$$Q = L \prod_{p \in P} p^{\lfloor \log_p H \rfloor}, \quad G = \left\lfloor \frac{H}{p_*} \right\rfloor.$$

For $a \in F$ the displacement term vanishes, since $a \mid Q$. For $a \notin F$ with $h(a) \leq H$, we have $h(a) \mid Q$ and hence $\langle Q, a \rangle \geq h(a)$. The complete-period term in (3) is therefore bounded by $h(a)/[a(b^{h(a)} - 1)]$, using the monotonicity of $n/(b^n - 1)$. The complete-period contributions therefore sum to less than ε . The dyadic estimate will control their incomplete periods.

For $h(a) > H$, we instead have $\langle Q, a \rangle \geq G$. Sum the complete-period bounds over $a \leq QT$, using $\sum_{a \leq QT} 1/a \leq 1 + \log(QT)$. There are at most QT incomplete-period terms, each at most $1/[T(b^G - 1)]$. The total contribution from these exponents is thus at most

$$\frac{G(1 + \log(QT)) + Q}{b^G - 1}.$$

For the gcd claim, either every P -prime-power component of $h(a)$ is at most H , in which case $h(a) \mid Q$, or some $p^{v_p(a)} > H$ contributes $p^{\lfloor \log_p H \rfloor} > H/p \geq H/p_* \geq G$ to the gcd. We can now choose the number M of dyadic lengths. The incomplete periods will cost a multiple of Q/M , whereas the preceding bound contains a multiple of GM/b^G . Thus M must lie between the polynomial scale of Q and the exponential scale b^G/G . With F and L already fixed, $Q \leq LH^{|P|}$ and $G = H/p_* + O(1)$, so we take $M = \lfloor b^{G/2} \rfloor$. The actual observation lengths are $T = 2^j$, with $M \leq j < 2M$. Combining the preceding estimates with (4), averaging over $T = 2^j$ for $M \leq j < 2M$, and using (5) with $\alpha_a = 1_A(a)/(b^{h(a)} - 1)$ yields

$$\frac{1}{M} \sum_{j=M}^{2M-1} \frac{1}{2^j} \sum_{t=1}^{2^j} \Delta_{b,A}(tQ) \leq \varepsilon + \frac{2QW_{b,P}(A)}{M} + \frac{G(1 + \log Q + 2M \log 2) + Q}{b^G - 1} + 4 \cdot 2^{-M}. \quad (6)$$

With F and L fixed and Q, G, M chosen as above, every term after ε tends to zero as $H \rightarrow \infty$. For sufficiently large H , the finite average is less than 2ε , and one sampled displacement is therefore less than 2ε . This selection gives no asserted rate of decay in the observation index $N = Qt$. Since ε is arbitrary, this contradicts the lower bound $1/q$ under rationality. Finally $W_{b,P}(A) \leq W_{2,P}(A)$ for $b \geq 2$, and weighted mass decreases on taking subsets. $\square$

2.1 An example beyond reciprocal summability

For the support in the introduction, the two sums can be compared layer by layer. Recall that

$$A_* = \{2^k m : k \geq 1, m \text{ odd}, m \leq 2^{2^k}\}.$$

The layers are disjoint, since their elements have different 2-adic valuations. For $r \geq 2$, put

$$S_r = \sum_{\substack{m \leq 2^r \\ m \text{ odd}}} \frac{1}{m}.$$

Each block $2^j \leq m < 2^{j+1}$, $1 \leq j < r$, contains 2^{j-1} odd integers, with reciprocal sum between $1/4$ and $1/2$. Including $m = 1$ gives $r/4 \leq S_r \leq r$. The ordinary reciprocal mass of the k th layer is $2^{-k} S_{2^k} \geq 1/4$, so $\sum_{a \in A_*} 1/a$ diverges. For $P = \{2\}$, that layer's weighted mass satisfies

$$\frac{S_{2^k}}{2^{2^k} - 1} \leq \frac{2^k}{2^{2^k} - 1} \leq 2^{1-k}.$$

The last bound uses $2^{2^k} - 1 \geq 2^{2^k-1}$ and $2^k \geq 2k$. Summation proves (W). The lower bound $1/4$ in every layer has already excluded reciprocal summability.

A related selection of a term from an average over an arithmetic progression occurs in Duverney–Tachiya [1, Section 2, (2.3)–(2.9)]. The sparse-coefficient criteria of Kaneko–Suzuki–Tachiya [2, Theorems 1 and 3] do not apply directly to c_A : for nonempty A , it is positive on every multiple of $\min A$. The corresponding density calculation, and the distinction between their remote-tail average and our displacement, are given in [Section 1.2 of the long record](#).

The dependence on the finite prime witness can be prescribed. If E is a finite set of primes and $\mathcal{U}$ is an upward-closed family of subsets of E with $E \in \mathcal{U} \not\supset \emptyset$, a finite union of prime-cofactor constructions has divergent reciprocal sum and $W_{b,P} < \infty$ exactly when $P \cap E \in \mathcal{U}$, for every $b \geq 2$ and finite prime set P , so every infinite subset is irrational at every base. The construction is given in [Section 1.3 of the long record](#).

3 Divisor covers and a common average

The weighted proof chose both its modulus and its observation window. To add a divisor-covered support, we need an estimate that is valid on that same window. We prove such a bound first, retaining its dependence on the modulus. This will allow the finite part of a cover to be added to the divisibility requirements on Q .

For $1 < B \leq 2$, positive integers L, d, M , and an integer $R \geq 0$, put

$$w_{B,d}(n) = \frac{B^{n \bmod d}}{B^d - 1}, \quad \mathcal{D}_{L;R,M} F = \frac{1}{M} \sum_{j=R}^{R+M-1} \frac{1}{2^j} \sum_{m=1}^{2^j} F(Lm).$$

The finite estimate

$$\mathcal{D}_{L;R,M} w_{B,d} \leq \frac{1 + 4L/M}{d(B-1)} \tag{S}$$

equivalently bounds $(B-1)\mathcal{D}_{L;R,M} w_{B,d}$ by $(1 + 4L/M)/d$. This normalised bound is uniform as $B \downarrow 1$; the unnormalised right side grows like $(B-1)^{-1}$. The ratio L/M measures the cost of incomplete modular periods.

Fix $T = 2^j$. We divide the proof according to how d compares with LT , which is the largest sampled multiple. When $d \leq LT$, the complete orbit has length d/g , with

$g = (L, d)$, and total weight $1/(B^g - 1)$. Complete cycles and one remaining piece give

$$\frac{1}{T} \sum_{m=1}^T w_{B,d}(Lm) \leq \frac{g}{d(B^g - 1)} + \frac{1}{T(B^g - 1)} \leq \frac{1}{d(B - 1)} + \frac{1}{T(B - 1)}.$$

Across dyadic lengths satisfying $d \leq L2^j$, the reciprocal-length errors sum to at most $2L/[d(B - 1)]$. When $d > 2LT$, there is no wrap and $2Lm \leq d - 1$; hence $\sum_{i=0}^{d-1} B^i \geq dB^{(d-1)/2} \geq dB^{Lm}$. Each atom is then at most $1/[d(B - 1)]$. Only the transition range $LT < d \leq 2LT$ remains. It contains at most one dyadic length, for which the geometric sum and convexity give

$$\frac{1}{T} \sum_{m=1}^T w_{B,d}(Lm) = \frac{B^L}{T(B^L - 1)} \frac{B^{LT} - 1}{B^d - 1} \leq \frac{LB^L}{d(B^L - 1)} \leq \frac{2L}{d(B - 1)}.$$

For the last inequality we used $(B^L - 1)/(B - 1) = \sum_{i=0}^{L-1} B^i \geq B^{L-1}$ and $B \leq 2$. Summing the main terms and the two error bounds proves (S). The corresponding [source estimate](#) uses the same finite average. Nonnegative interchange permits summation against any coefficients c_d with $\sum_d c_d/d < \infty$.

3.1 Positive divisor majorants

For a finite set F , write $f_F(n) = \#\{a \in F : a \mid n\}$. A positive divisor sum is convenient because each divisor contributes one copy of the kernel just estimated. We therefore look for a bound on f_F^α , with $0 < \alpha \leq 1$, by such a sum. Varying α from one finite set to another can make the resulting costs summable.

Theorem 3.1 (a summable divisor-cover criterion). *For each $j \geq 1$, let $F_j \subseteq \mathbb{N}_{>0}$ be finite, let $0 < \alpha_j \leq 1$, and let $c_{j,d} \geq 0$ satisfy*

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$$f_{F_j}(n)^{\alpha_j} \leq \sum_{d \mid n} c_{j,d} \quad (n \geq 1).$$

Set $C_j = \sum_{d \geq 1} c_{j,d}/d$. If

$$\sum_{j \geq 1} \frac{C_j 2^{j\alpha_j}}{2^{\alpha_j} - 1} < \infty, \tag{V}$$

then $X_A(b)$ is irrational for every infinite $A \subseteq \bigcup_j F_j$ and every integer $b \geq 2$.

For example, take $F_j = \{4^j\}$, $\alpha_j = 1$ and $c_{j,4^j} = 1$, with all other coefficients zero. Then $C_j = 4^{-j}$, and the cost in (V) is $\sum_j 2^{-j}$. A majorant exists for any finite F_j : the choice $\alpha_j = 1$ and $c_{j,d} = \mathbf{1}_{F_j}(d)$ suffices. What restricts an infinite support is summability of the entire sequence of costs. Section 4 gives a necessary condition that applies to every possible cover.

Proof. Write $B_j = 2^{\alpha_j}$ and

$$U_j(N) = \sum_{r \geq 1} 2^{-r} f_{F_j}(N + r), \quad V_j(N) = \sum_{d \geq 1} c_{j,d} w_{B_j,d}(N).$$

We first pass from divisor counts to the shifted tails U_j . The power α_j changes their geometric factor to B_j^{-r} , so subadditivity and the assumed majorant yield

$$U_j(N)^{\alpha_j} \leq \sum_{r \geq 1} B_j^{-r} f_{F_j}(N+r)^{\alpha_j} \leq \sum_{d \geq 1} c_{j,d} \sum_{\substack{r \geq 1 \\ d|N+r}} B_j^{-r} = V_j(N).$$

Summing the progression in each residue class gives the last equality. For the displacement itself, the geometric-series identity gives

$$\Delta_{2,F_j}(N) = U_j(N) - X_{F_j}(2) \leq U_j(N).$$

Consequently, once the first J finite sets have zero displacement, nonnegativity bounds the displacement of their union by $\sum_{j>J} U_j(N)$.

Fix $\varepsilon > 0$, set $t_j = \varepsilon 2^{-j}$, and choose J with

$$K_J := \sum_{j>J} \frac{C_j t_j^{-\alpha_j}}{B_j - 1} < \frac{1}{4}.$$

This is possible since $\varepsilon^{-\alpha_j} \leq \max(1, \varepsilon^{-1})$. Choose L divisible by every member of the first J finite sets. Equation (S) bounds the finite mean of $S_J(N) := \sum_{j>J} t_j^{-\alpha_j} V_j(N)$ by $(1 + 4L/M)K_J < 1/2$ whenever $M \geq 4L$. We choose a sample with $S_J(N) < 1$. Every summand is nonnegative, so this one choice gives $U_j(N) < t_j$ for all $j > J$ simultaneously. Every exponent in the first J finite sets divides L , so those terms have zero displacement. Consequently,

$$0 < \Delta_{2,A}(N) \leq \sum_{j>J} U_j(N) \leq \varepsilon.$$

Equation (D) now excludes rationality at base two. For $0 \leq r < d$, the function $(b^r - 1)/(b^d - 1)$ is nonincreasing on $b > 1$. For $r > 0$, cancel $b - 1$ and write it as $A(b)/(A(b) + C(b))$, where $A(b) = \sum_{i<r} b^i$ and $C(b) = \sum_{r \leq j < d} b^j$. Its derivative has the sign of $A'C - AC' = \sum_{i<r \leq j < d} (i-j)b^{i+j-1} < 0$. Thus $\Delta_{b,A}(N) \leq \Delta_{2,A}(N)$ for every real $b \geq 2$; when b is an integer, the same rational-lattice argument applies. Any prescribed positive divisor can be included in L , so the witnesses can also be required to be arbitrarily large. $\square$

The same proof permits any positive weights η_j with $\sum_j \eta_j = 1$: replace $2^{j\alpha_j}$ in (V) by $\eta_j^{-\alpha_j}$ and take $t_j = \varepsilon \eta_j$. The stronger condition $\sum_j C_j 2^{j\alpha_j} / (2^{\alpha_j} - 1)^2 < \infty$ implies (V), since each of its summands is the corresponding summand of (V) multiplied by $2^{\alpha_j} / (2^{\alpha_j} - 1) > 1$. This proves inclusion of the hypotheses for a given cover; after optimisation over all covers, the inclusion of support classes is strict by a [corresponding source theorem](#).

3.2 Combining the two support criteria

For a union, the weighted displacement and the cover tail must be small at the same integer. We arrange this by including both finite sets of divisibility requirements in the initial modulus, then applying both estimates to the distribution chosen in the weighted proof.

Theorem 3.2 (mixed weighted and cover supports). *Let $E, V \subseteq \mathbb{N}_{>0}$. Suppose E satisfies (W) for a finite nonempty prime set P , and $V \subseteq \bigcup_j F_j$ for finite sets and nonnegative majorants satisfying the hypotheses of Theorem 3.1, with either (V) or its positive-weight variant. Then $X_A(b)$ is irrational for every infinite $A \subseteq E \cup V$ and every integer $b \geq 2$.*

Proof. Fix $\varepsilon > 0$ and an integer $N_0 \geq 1$, and put $\rho = \varepsilon/3$. Use the cover notation B_j, U_j, V_j above, with $\eta_j = 2^{-j}$ in the case (V). Set $t_j = \rho\eta_j$ and choose J so that

$$K_J = \sum_{j>J} \frac{C_j t_j^{-\alpha_j}}{B_j - 1} < \frac{1}{16}.$$

Choose a finite $F \subseteq E$ whose complementary weighted mass is $\kappa < \rho/16$. Let L be a positive common multiple of N_0 , all members of F , and all members of the first J finite sets. Keeping J, F and L fixed, we use the base-two choices

$$Q = L \prod_{p \in P} p^{\lfloor \log_p H \rfloor}, \quad G = \lfloor H / \max P \rfloor, \quad M = \lfloor 2^{G/2} \rfloor.$$

The proof of (6) allows any multiple of F as L , including the extra divisors imposed by the cover. The estimate also holds for finite or empty E , since infinitude was used only to make the final displacement positive. Consequently

$$\mathcal{D}_{Q;M,M}(\Delta_{2,E}/\rho) < \frac{1}{8}$$

for sufficiently large H . For the same finite distribution, (S) and nonnegative interchange give

$$\mathcal{D}_{Q;M,M} S_J \leq (1 + 4Q/M) K_J < \frac{1}{8}, \quad S_J = \sum_{j>J} t_j^{-\alpha_j} V_j,$$

because $Q/M \rightarrow 0$. Adding the two estimates bounds the mean of $\Delta_{2,E}/\rho + S_J$ by $1/4$. We choose a sample $N = Qm$ at which this sum is less than 1. Positivity then gives both $\Delta_{2,E}(N) < \rho$ and $U_j(N) < t_j$ for every $j > J$. The displacement terms from the first J finite sets vanish because their exponents divide L , and hence divide N . Therefore

$$\Delta_{2,A}(N) \leq \Delta_{2,E}(N) + \Delta_{2,V}(N) < 2\rho < \varepsilon, \quad N \geq Q \geq L \geq N_0.$$

This inequality remains valid when E and V overlap. Since A is infinite, its displacement is positive. The comparison of atoms in the cover proof gives arbitrarily small positive displacements at every integer base, and (D) finishes the proof. $\square$

4 Comparison of the support classes

We shall construct E satisfying only the weighted condition and V satisfying only the cover condition. To rule out every cover of E , we first derive a lower bound that depends on its finite subsets rather than on a choice of majorants. For finite F , let $\mathbb{E}_F$ denote the

uniform mean modulo $\text{lcm}(F)$, with $\text{lcm}(\emptyset) = 1$ and $f_\emptyset = 0$. More generally allow weights $\eta_j > 0$ with $\sum_j \eta_j = 1$, and set

$$K = \sum_j \frac{C_j \eta_j^{-\alpha_j}}{2^{\alpha_j} - 1}, \quad \Psi(0) = 0, \quad \Psi(t) = \inf_{0 < \alpha \leq 1} \frac{t^\alpha}{2^\alpha - 1} \quad (t \geq 1).$$

Here $\log^+ t = \log \max\{1, t\}$. For every finite F contained in $\bigcup_j F_j$,

$$K \geq \mathbb{E}_F \Psi(f_F) \geq e \mathbb{E}_F \log^+ f_F. \quad (7)$$

Indeed, at a point where $f_F(n) = t > 0$, some covering set F_j satisfies $f_{F_j}(n) \geq \eta_j t$. Otherwise summing contradicts coverage. The corresponding weighted majorant is at least $\Psi(t)$. Average first over $1 \leq n \leq X$, using $\lfloor X/d \rfloor / X \leq 1/d$, and let $X \rightarrow \infty$. The periodic left side tends to $\mathbb{E}_F \Psi(f_F)$, proving the first inequality; the covering moduli need not divide $\text{lcm}(F)$. Convexity gives $2^\alpha - 1 \leq \alpha$, and $e^u/u \geq e$ proves the second. The scalar inequality $t^\alpha / (2^\alpha - 1) \geq e \log t$ also has a [corresponding source theorem](#).

On a divisor cube the lower bound is asymptotically attained. For $F(q, P) = \{qd : d \mid \prod_{p \in P} p\}$, where $q \geq 2$ and no $p \in P$ divides q , put $S = \sum_{p \in P} 1/p$. If $S \geq 1$, the infimum K_* over finite or countable covers satisfies

$$\frac{e(S-1)}{q} \leq K_*(F(q, P)) \leq \frac{eS}{q}. \quad (8)$$

For the lower bound, we condition on $q \mid n$ and write $f_F(n) = 2^z$. The Chinese remainder theorem gives $\mathbb{E}Z = S$. For $z \geq 0$ and $v = \alpha \log 2 > 0$,

$$\frac{2^{\alpha z}}{2^\alpha - 1} \geq \frac{e^{(z-1)v}}{v} \geq e(z-1) \quad (z > 1);$$

for $0 \leq z \leq 1$ the claimed lower bound is nonpositive. Thus $\Psi(2^z) \geq e(z-1)$. For the upper bound, use the one-set cover with $z = 1/S$ and $\alpha = \log_2(1+z)$; its exact positive expansion has cost

$$\frac{1}{q^z} \prod_{p \in P} (1 + z/p) \leq \frac{eS}{q}.$$

In particular, $1 - 1/S \leq qK_*(F(q, P))/(eS) \leq 1$. Thus $K_*(F(q, P)) \sim eS/q$ as $S \rightarrow \infty$, uniformly over the permitted choices of q and P .

The logarithmic lower bound is not a converse: replacing a fractional majorant by a logarithmic one need not control averages along the multiples of a prescribed modulus. The counterexample and the bound that remains valid on ordinary initial intervals are proved in [Section 13 of the long record](#); those proofs were first worked out in an AI-assisted note of 17 September 2026. These are statements about finite averages; they do not rule out an irrationality criterion based on other uses of logarithmic divisor counts.

We apply (7) to successive finite unions of divisor cubes to obtain the required separating supports. Write $\mathcal{W}_b$ for the supports satisfying the finite-prime weighted hypothesis at base b , and $\mathcal{C}$ for supports admitting a strengthened positive cover as in (V).

Proposition 4.1 (incomparable support criteria). *There are infinite positive supports E and V such that*

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$$E \in \mathcal{W}_2, \quad E \notin \mathcal{C}, \quad V \in \mathcal{C}, \quad V \notin \mathcal{W}_b \ (b \geq 2), \quad \sum_{a \in V} \frac{1}{a} = \infty.$$

Proof. The reciprocal sum over odd primes diverges, even after finitely many are removed. Otherwise $\prod_{p \text{ odd}} (1 + 1/p)$ would bound the reciprocal mass of odd squarefree integers; the decomposition $n = ds^2$ would then make the odd harmonic series converge.

Choose pairwise disjoint finite sets P_k of odd primes such that $2^k/4 \leq S_k := \sum_{p \in P_k} 1/p < 2^k/4 + 1$. Put $M_k = \prod_{p \in P_k} p$ and $E = \bigcup_{k \geq 1} \{2^k d : d \mid M_k\}$. The witness $\{2\}$ has weighted mass

$$\sum_{k \geq 1} \frac{\prod_{p \in P_k} (1 + 1/p)}{2^{2^k} - 1} \leq 2e \sum_{k \geq 1} \exp(-(\log 2 - 1/4)2^k) < \infty.$$

For the union F_m of the first m frames, average $\log^+ f_{F_m}$ over one period. On $v_2(n) = k < m$, the k th frame contributes $2^{Z_k(n)}$ divisors, where $Z_k(n) = \sum_{p \in P_k} \mathbf{1}_{p|n}$. This event has density $2^{-(k+1)}$ and is independent of odd-prime divisibility. Thus

$$\mathbb{E}_{F_m} \log^+ f_{F_m} \geq (\log 2) \sum_{k=1}^{m-1} 2^{-(k+1)} S_k \geq \frac{m-1}{8} \log 2.$$

Bound (7) makes these means uniformly bounded for every strengthened cover, so $E \notin \mathcal{C}$.

For the reverse direction, choose unused odd primes $q_j \geq 4^j$ and disjoint finite sets P_j of unused odd primes. Stop each P_j at the first prefix with $R_j := \prod_{p \in P_j} (1 + 1/p) \geq q_j$. Prime reciprocal divergence permits this, and minimality gives $q_j \leq R_j < 4q_j/3$. Put $S_j = \sum_{p \in P_j} 1/p$, $M_j = \prod_{p \in P_j} p$, and $V = \bigcup_{j \geq 1} F_j$, where $F_j = \{q_j d : d \mid M_j\}$. The frames have disjoint prime alphabets, and

$$\sum_{a \in F_j} \frac{1}{a} = \frac{R_j}{q_j} \in [1, 4/3).$$

Thus the reciprocal sum diverges. Every finite prime set misses all factors of all but finitely many frames; each missed frame has base- b weighted mass $R_j/[q_j(b-1)] \geq 1/(b-1)$. Hence $V \notin \mathcal{W}_b$ for every integer $b \geq 2$.

For the full infinite-cover cost, from $\log(1+x) \geq x - x^2/2$ and $\sum_{p \in P_j} p^{-2} \leq 1$ we have $S_j \leq \log R_j + 1/2 < \log q_j + 1$. Set

$$z_j = (j + S_j)^{-1}, \quad \alpha_j = \log_2(1 + z_j), \quad c_{j,q_j d} = z_j^{\omega(d)} \quad (d \mid M_j),$$

with other coefficients zero and $\omega(d)$ the number of prime factors of d . If $q_j \mid n$ and Z primes in P_j divide n , the divisor majorant equals $(1 + z_j)^Z = f_{F_j}(n)^{\alpha_j}$; otherwise both sides vanish. The column cost is $C_j = q_j^{-1} \prod_{p \in P_j} (1 + z_j/p)$, whence

$$\frac{C_j 2^{j\alpha_j}}{2^{\alpha_j} - 1} = \frac{(1 + z_j)^j}{q_j z_j} \prod_{p \in P_j} (1 + z_j/p) \leq \frac{e(j + S_j)}{q_j} \leq e((1 + \log 4)j + 1)4^{-j}.$$

The last bound uses that $(j + \log x + 1)/x$ decreases for $x \geq 4^j$. Its sum converges, including the frame-index factor in (V); hence $V \in \mathcal{C}$. $\square$

4.1 A strict enlargement and finite unions

The union of the two separating supports requires the mixed theorem.

Corollary 4.2 (a host requiring the mixed criterion). *There is an infinite positive support U with $U \notin \mathcal{C}$ and $U \notin \mathcal{W}_b$ for every integer $b \geq 2$, such that $X_A(b)$ is irrational for every infinite $A \subseteq U$ and every integer $b \geq 2$.*

Proof. Let $U = E \cup V$, with E, V as in Proposition 4.1. A cover of U restricts to a cover of E , and a finite-prime weighted witness for U restricts to one for V , since all weights are nonnegative. Thus U belongs to neither individual class. Theorem 3.2 gives irrationality for every infinite subset of U at every integer base $b \geq 2$. $\square$

The fixed-dyadic cover class in (V) is closed under finite unions. For two covers, interleave their j th frames at positions $2j - 1, 2j$, halve each exponent, and retain the nonnegative coefficient columns. The majorants persist since $f^{\alpha/2} \leq f^\alpha$ for every nonnegative integer f . Writing $T_j = C_j 2^{j\alpha_j} / (2^{\alpha_j} - 1)$ for an old column, its new cost is at most

$$\frac{C_j 2^{j\alpha_j}}{2^{\alpha_j/2} - 1} = (1 + 2^{\alpha_j/2}) T_j \leq (1 + \sqrt{2}) T_j.$$

For r covers, interleave at positions at most rj and divide each exponent by r ; the cost ratio is $(2^\alpha - 1)/(2^{\alpha/r} - 1) < 2r$. Weighted supports are likewise closed under finite unions: combine their finite prime witnesses, so the prime part grows and $h/(b^h - 1)$ decreases. Finite supports admit both criteria; subsets inherit each host. Thus the mixed class is closed under finite unions and finite changes using the literal dyadic condition (V), without changing to arbitrary cover weights. The summability hypothesis cannot be omitted for countable unions. Every prime singleton is admitted, but for the full prime support $\mathcal{P}$ one has $\Delta_{2, \mathcal{P}}(N) > 1/3$ for every $N \geq 1$: indeed $\sum_{r \geq 1} 2^{-r} \omega(N+r) \geq 1$, whereas $X_{\mathcal{P}}(2) \leq \sum_{a \geq 2} (2^a - 1)^{-1} < 2/3$. Here $\omega(n)$ is the number of distinct prime divisors of n . The reciprocal-summable class is contained in the weighted class, since $h/(2^h - 1) \leq 1$. A direct averaging proof of this special case is given in Section A.

For comparison, Tao–Teräväinen prove the full-prime case at base 2 [3, Theorem 1.3, p. 4]. The paragraph following it states extensions to prime support at every integer base and to full prime-power support at base 2, leaving the modifications to the reader. It does not treat arbitrary infinite thinnings. The proposed thinning extension is not a premise of any theorem here.

5 Limits of the small-displacement criterion

The small-displacement condition does not cover even every support whose sum is already known to be irrational. For full support we have

$$\Delta_{2, \mathbb{N}_{>0}}(N) > (2^N - 1) \sum_{a > N} 2^{-a} = 1 - 2^{-N} \geq \frac{1}{2} \quad (N \geq 1).$$

The full-support value is nevertheless [irrational](#) [5]. Thus a proof for arbitrary support would have to use more than this particular family of small displacements. The example leaves other integer linear forms and averaging arguments available.

For squarefree support, Duverney and Tachiya's theorem already gives irrationality. It holds at every base 2^j , $j \geq 1$, by Corollary 1.2 and Example 1.1 of Duverney and Tachiya [1, p. 4]. Their corollary covers the s -free products of any pairwise coprime sequence of polynomial growth, and they present it as support for the conjecture of Erdős and Graham stated here as Problem 1.1. For the squarefree support with 1 removed, the divisor count is $2^{\omega(n)} - 1$, which is odd for $n \geq 2$. This excludes a first-block-divisibility condition at even bases. Adjoining 1 removes that parity obstruction while changing the sum by a rational number. The argument concerns that normalisation of the certificate, and gives no obstruction to irrationality of the sum. Details and the other known support families appear in [the comparison with known support theorems in the long record](#).

A growth-only criterion gives a different class: if $A = \{c_1 < c_2 < \dots\}$ and $\limsup_n c_n/2^n = \infty$, then $X_A(b)$ is irrational for every integer $b \geq 2$ [8, Theorem 1]. The same section of the long record proves the translation from denominator growth and gives examples showing that this class and reciprocal summability are incomparable. The interval-filling constructions in [9] and the freely chosen lacunary denominators of [16] are not constructions of subseries of these fixed Mersenne weights.

Prime-incidence arguments require additional care as well: multiplying all prime exponents by 2 creates positive correlations between their divisibility indicators. The identity and covariance calculation appear at the end of [the long record's discussion of mixed supports](#). Size conditions alone do not supply that independence.

5.1 Algebraic bases on separated supports

The companion [Reading eight Erdős problems together](#), subsection “Divisibility cuts at every algebraic base”, states a transcendence extension based on the number-field Subspace Theorem. Its hypothesis is an infinite host H with cuts $L_j < M_j$, where $L_j \rightarrow \infty$ and $M_j - L_j \rightarrow \infty$, such that every $n \in H$ with $n \leq L_j$ divides L_j , and every $n \in H$ with $n > L_j$ is a multiple of M_j . The claimed conclusion is transcendence of $\sum_{n \in B} w_n/(t^n - 1)$ for every infinite $B \subseteq H$, bounded positive integer weights w_n and real algebraic $t > 1$.

Every divisibility chain has such cuts, as does $H_* = \bigcup_j N_j \{1, \dots, 2^{N_j}\}$ with $N_0 = 1$ and $N_{j+1} = 2N_j \operatorname{lcm}(1, \dots, 2^{N_j})$. Each block has reciprocal mass at least $1/2$ and contains antichains of unbounded size. Its one-prime weighted mass at every real $t > 1$ is at most $2t^2/(t-1)^3$. This provides a connection with the weighted criterion, although the cut condition is not asserted for every weighted host.

The companion proof is outside this paper and is not a premise of the weighted, cover or mixed theorem. It has two AI proof reviews, no Lean proof of transcendence and no independent human review. No historical priority is asserted for the extension. The case of arbitrary infinite support at base two remains unresolved.

A Reciprocal-summable supports

Erdős stated the following extension of his pairwise-coprime theorem [7, p. 222]. It follows from Theorem 1.2, since $h/(2^h - 1) \leq 1$, but the direct argument explains why finite reciprocal mass makes the averaging simpler.

Theorem A.1 (reciprocal-summable supports). *Let $A \subseteq \mathbb{N}_{>0}$ be infinite. If*

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$$\sum_{a \in A} \frac{1}{a} < \infty,$$

then $X_A(b)$ is irrational for every integer $b \geq 2$.

Erdős also discussed weaker conditions [7, pp. 222, 226]. We do not identify our weighted hypothesis with the conditions he suggested.

Here we can take the observation length to infinity while holding the modulus fixed: reciprocal summability supplies a summable bound for all the remaining exponents. We then enlarge the modulus to make each fixed exponent divide it. No common period for the whole infinite support is required.

Fix the integer base $b \geq 2$ and put

$$w_{b,d}(N) = \frac{b^{N \bmod d}}{b^d - 1}, \quad T_N^{(b)} = \sum_{d \in A} w_{b,d}(N).$$

Then $T_N^{(b)} - T_0^{(b)} = \Delta_{b,A}(N)$. For a fixed positive integer Q , the orbit of Q modulo d consists of d/g residues, where $g = (Q, d)$. Summing the geometric progression gives

$$M_{Q,b}(d) := \lim_{X \rightarrow \infty} \frac{1}{X} \sum_{m=1}^X w_{b,d}(Qm) = \frac{g}{d(b^g - 1)} \leq \frac{1}{d}. \quad (9)$$

The orbit 4, 2, 0 modulo 6 used in Section 2 illustrates this mean: its atom average is $1/9$, while its displacement average is $2/21$. When an exponent divides Q , its atom stays at its zero-shift value, so its contribution to the displacement vanishes.

To interchange the sum over exponents and this limit, we use a bound summable in d for the fixed modulus Q . Counting positive multiples of d gives

$$\begin{aligned} \frac{1}{X} \sum_{m=1}^X w_{b,d}(Qm) &= \sum_{r \geq 1} b^{-r} \frac{\#\{1 \leq m \leq X : d \mid Qm + r\}}{X} \\ &\leq \sum_{r \geq 1} b^{-r} \frac{Q+r}{d} = \frac{Q/(b-1) + b/(b-1)^2}{d} \leq \frac{Q+2}{d}. \end{aligned}$$

Indeed, the counted multiples are distinct and at most $QX + r$. For this fixed Q , the majorant $(Q+2)/d$ is summable over A . Dominated convergence therefore takes the observation-length limit inside the sum over d , giving the Cesàro mean $\sum_{d \in A} M_{Q,b}(d)$. This majorant is not uniform in growing Q .

Next let $Q_t = \text{lcm}(1, \dots, t)$. For each fixed d , eventually $d \mid Q_t$ and $M_{Q_t, b}(d) = (b^d - 1)^{-1}$. A second dominated-convergence argument, using $M_{Q_t, b}(d) \leq 1/d$, yields

$$\sum_{d \in A} M_{Q_t, b}(d) \rightarrow X_A(b). \quad (10)$$

Thus the limiting averages of the nonnegative displacements $\Delta_{b, A}(Q_t m)$ tend to zero. Choose t , then a sufficiently long finite average, and finally a term no larger than that average. This gives arbitrarily small positive displacements, contradicting the rational lattice in (D). This proves Theorem A.1 directly at every integer base.

The order of limits is essential: the observation length tends to infinity with the modulus fixed, and only then does the modulus increase. Reciprocal summability controls both interchanges.

For example, every powerful integer (each prime factor occurs to exponent at least two) has the form $u^2 v^3$. Hence its reciprocal sum is at most $\zeta(2)\zeta(3)$, and the theorem applies to every infinite subset of the powerful integers. For full perfect-power supports, compare Duverney–Tachiya [1, Corollary 1.2, p. 4] and the earlier linear-independence results of Luca–Tachiya [13].

B Finite-support denominator periods

For a finite nonempty $F \subseteq \mathbb{N}_{>0}$, let D_F be the positive reduced denominator of $X_F(b) = \sum_{n \in F} (b^n - 1)^{-1}$. We use $\text{ord}_1(b) = 1$.

Theorem B.1 (the exact denominator period). *Let $F \subseteq \mathbb{N}_{>0}$ be finite and nonempty, let $b \geq 2$ be an integer, and let $D_F > 0$ be the denominator of $X_F(b)$ in lowest terms. Then D_F is coprime to b , and*

$$\text{ord}_{D_F}(b) = \text{lcm}\{n : n \in F\}.$$

If moreover $\text{lcm}(F) \geq 2$, then $\text{lcm}(F) < D_F$. We use $\text{ord}_1(b) = 1$, so the statement includes $F = \{1\}$ at $b = 2$.

Put $L = \text{lcm}(F)$. Clearing denominators gives $D_F \mid b^L - 1$, so the upper divisibility for the order is immediate. For the reverse, choose $n \geq 2$ maximal under divisibility in F and a prime $\ell \mid \Phi_n(b)$. If $e = v_\ell(b^n - 1)$, the full prime power ℓ^e has $\text{ord}_{\ell^e}(b) = n$. Every other selected exponent m has $n \nmid m$, hence $v_\ell(b^m - 1) < e$. The n th summand has uniquely smallest ℓ -adic valuation and cannot cancel. Thus $\ell^e \mid D_F$ and $n \mid \text{ord}_{D_F}(b)$. Taking all maximal selected exponents proves the order statement; the size bound follows from $\text{ord}_{D_F}(b) \mid \varphi(D_F) < D_F$ when $L \geq 2$. The case $F = \{1\}$ has order one directly.

The unique-valuation argument also permits signs ± 1 on the finite summands. This signed extension is not included in the formal statement. The required cyclotomic prime-power fact is proved in Section E. The distinction between primes and prime powers is visible in

$$X_{\{2,3\}}(2) = \frac{10}{21}, \quad X_{\{2,6\}}(2) = \frac{22}{63}.$$

In both examples 2 has multiplicative order six modulo the denominator. The first combines orders two and three; the second retains the order-six prime power 9, although

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no prime divisor of 63 has order six. The boundary $F = \{1\}$ at base two has $D_F = L = 1$. These lower denominator bounds give no upper height control for infinite partial sums and do not decide an infinite-support value. Van Assche instead constructs approximants for the full Lambert series and proves both nonvanishing and decay of the associated integer linear forms [14, Lemma 1 and (34)–(35)]. That approximation mechanism does not follow from denominator survival alone.

In particular, $1/21$ is not a finite subseries sum. If it were, the order of 2 modulo 21 would force every selected exponent to divide 6. The exponent 1 is excluded since its weight is 1. Any sum containing 2 or 3 exceeds $1/21$, whereas the only nonzero remaining sum is $1/63$, from the exponent 6. This will distinguish finite and infinite representations in the next appendix.

C Rational membership

C.1 Rational values and scaled tails

Suppose $X_A(2) = p/v$, where $p \in \mathbb{Z}$ and $v \geq 1$ is an integer. Multiplying the coefficient-series identity by $v2^N$ gives

$$z_N := v \sum_{r \geq 1} c_A(N+r)2^{-r} = 2^N p - v \sum_{n=1}^N c_A(n)2^{N-n} \in \mathbb{Z}.$$

Separating the first term of the tail then gives $z_{N+1} = 2z_N - vc_A(N+1)$. All the coefficients $c_A(n)$ are divisor counts of the same set A .

Two elementary tail properties hold regardless of whether the value is rational. Assume first that A is nonempty and put $a_0 = \min A$. Then every a_0 consecutive integers contain a multiple of a_0 , so a divisor-incidence zero window has length at most $a_0 - 1$. If A is infinite, choose k exponents and a common multiple L ; then $c_A(L) \geq k$ and $\sum_{r \geq 1} c_A(L-1+r)2^{-r} \geq k/2$, by its first term. Both facts hold without rationality. A contradiction must use the integral recurrence together with the requirement that every coefficient is obtained from the same selector.

The compatibility condition on the coefficients is expressed by Dirichlet convolution: $\mu * c_A = \mathbf{1}_A$, where μ is the Möbius function. A putative recurrence with integral forcing must therefore satisfy $(\mu * c_A)(n) \in \{0, 1\}$ for all n . The telescoping proof of the integer-recurrence criterion and the Möbius inversion argument are given under [Bounds for a general coefficient sequence](#) in the long record. The example $A = \{2\}$ there has value $1/3$. The correspondence permits finite supports; infinitude remains a separate requirement for a counterexample to Problem 1.1.

C.2 The achievement set

Let $w_n = (2^n - 1)^{-1}$ and

$$A = \left\{ \sum_{n \geq 1} \varepsilon_n w_n : \varepsilon_n \in \{0, 1\} \right\}, \quad R_N = \sum_{n > N} w_n.$$

The estimate

$$2^{-N} < R_N \leq 2^{-N} + \frac{2}{3} 4^{-N} < w_N \quad (N \geq 1)$$

implies uniqueness of the selector. Indeed, at the first differing exponent n , the difference w_n exceeds everything later terms can cancel. The middle inequality follows by writing $w_n = 2^{-n} + 4^{-n}/(1 - 2^{-n})$ and using $n \geq N + 1 \geq 2$. Since every weight exceeds its tail, Hornich's theorem [15], as proved by Nitecki [10, Theorem 4(1), p. 9], shows that the coding image is a Cantor set of measure $\lim_N 2^N R_N$; the estimate above gives $2^N R_N \rightarrow 1$, so $\lambda(A) = 1$. Concretely, fixing the first N digits gives 2^N disjoint closed intervals of length R_N . These interval unions decrease to A , and continuity of Lebesgue measure from above gives the same limit. Kovač–Tao record this strict-tail inequality and the Cantor conclusion in the fixed-base setting [4, Remark 4.1]. Thus each represented target has a unique selector. The [geometry section of the long record](#) also treats restricted supports. The linked results give the [comparison of each restricted tail with the preceding weight](#), the [injectivity of the restricted digit map](#) and the [measure dichotomy](#).

C.3 The greedy recurrence

For $x \geq 0$, the greedy rule starts with $r_0 = x$ and, at rank $n \geq 1$, selects n when $r_{n-1} \geq (2^n - 1)^{-1}$, subtracting that weight if selected. Let A_x be the resulting support and $c_x(n) = \#\{a \in A_x : a \mid n\}$. Set

$$P_0 = 0, \quad P_{N+1} = 2P_N + c_x(N + 1), \quad Q_N = \lfloor 2^N x \rfloor - P_N.$$

Thus $P_N = \sum_{j=1}^N 2^{N-j} c_x(j)$ is the integer truncation of the Lambert expansion (2). It is at most $2^N X_{A_x}(2) \leq 2^N x$, so $Q_N \geq 0$. These integer remainders obey

$$Q_{N+1} = 2Q_N + \beta_N - c_x(N + 1), \quad \beta_N = \lfloor 2^{N+1} x \rfloor - 2\lfloor 2^N x \rfloor \in \{0, 1\}. \quad (11)$$

Theorem C.1 (membership and nonincreasing integer remainders). *For every real $x \geq 0$, the following are equivalent:*

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$$\begin{aligned} & x \in A; \\ & \forall K \geq 0 \exists N \geq K : Q_{N+1} \leq Q_N; \\ & \forall K \geq 0 \exists N \in [K, K + 2\lfloor \sqrt{K} \rfloor + 12) : Q_{N+1} \leq Q_N. \end{aligned}$$

Proof. Strict domination of each Mersenne weight over its tail implies that a represented target is recovered by the greedy rule. For such a target,

$$0 \leq Q_N \leq \sum_{r \geq 1} c_x(N + r) 2^{-r} \leq 2\sqrt{N} + 4,$$

since $c_x(n) \leq \tau(n) \leq 2\sqrt{n}$ and $\sqrt{N+r} \leq \sqrt{N} + \sqrt{r} \leq \sqrt{N} + r$. Put $s = \lfloor \sqrt{K} \rfloor$ and $T = 2s + 12$. Strict increase at all T steps would imply $Q_{K+T} \geq T$. But $K + T < (s + 4)^2$ gives $Q_{K+T} < 2s + 12 = T$. This proves the window condition and hence cofinal nonincreases.

Conversely, if $x \notin A$, then $\delta = x - X_{A_x}(2) > 0$. The Lambert prefix is at most $2^N X_{A_x}(2)$, so $Q_N \geq 2^N \delta - 1$. Eventually this exceeds $c_x(N + 1) \leq N + 1$. Equation (11) then makes Q_N strictly increasing, contradicting cofinal nonincreases. $\square$

The theorem supplies a test for membership. Applying it requires nonincreases along the actual greedy sequence of the target. For $x = 0$, the selector is empty and $Q_N = 0$ at every rank. A finite subseries sum is likewise represented and satisfies the criterion. The target $x = 3/4$ lies strictly between $R_1 < 2/3$ and $w_1 = 1$, so it is not represented and its integer remainders eventually increase strictly. For $1/2$ and $1/21$, the theorem does not establish which alternative occurs.

The same argument allows a uniform subpower window: for each $0 < \varepsilon < 1$, a constant C_ε gives the window length $\lceil C_\varepsilon(K+1)^\varepsilon \rceil$. Indeed, $\tau(n) \leq A_\varepsilon n^\varepsilon$ gives $Q_N \leq D_\varepsilon(N+1)^\varepsilon$ for represented targets, where $D_\varepsilon = A_\varepsilon \sum_{r \geq 1} r^\varepsilon 2^{-r}$. Choose $C_\varepsilon > D_\varepsilon(C_\varepsilon+3)^\varepsilon$, which is possible because $\varepsilon < 1$. For $T = \lceil C_\varepsilon(K+1)^\varepsilon \rceil$, strict increase at all T steps would give

$$T \leq Q_{K+T} \leq D_\varepsilon(K+T+1)^\varepsilon \leq D_\varepsilon(C_\varepsilon+3)^\varepsilon(K+1)^\varepsilon < T,$$

a contradiction. The converse still follows from exponential growth when x is not represented. This shortens the window for represented targets without deciding whether a specified target is represented. The linked formal proof establishes the square-root window.

At $x = 1/2$, the digits β_N vanish for $N \geq 1$; at $x = 1/21$ they are six-periodic. Thus the unresolved arithmetic input is

$$\forall K \exists N \geq K : \quad c_x(N+1) \geq Q_N + \beta_N, \quad x \in \{1/2, 1/21\}. \quad (12)$$

Both Q_N and c_x must arise from the same greedy selector. Counterexamples to fixed-multiplier schedules, with their finite phase masks, are retained in the long record. They concern those schedules; they do not decide either target.

D Integer quotients, approximation and further questions

For $1/21$ an integer greedy procedure gives finite remainders that can be compared with the real greedy sequence. For $1/2$ we instead allow arbitrary finite approximating supports. In both cases the outstanding question is whether suitable finite data occur at unbounded depths.

Integer quotients for $1/21$. For $M, d \geq 1$, set

$$q_M(d) = \left\lfloor \frac{2^M}{2^d - 1} \right\rfloor, \quad T_M = \left\lfloor \frac{2^M}{21} \right\rfloor.$$

Starting with the integer remainder T_{2R} , consider $d = 2, \dots, R$ in order and subtract $q_{2R}(d)$ whenever it does not exceed the current remainder. Let D_R be the set of selected exponents and s_R the final remainder. Thus

$$s_R = T_{2R} - \sum_{d \in D_R} q_{2R}(d) \geq 0.$$

A second comparison uses the same integer rule through $d = 2R$, rather than stopping at R . For example, $R = 6$ gives $T_{12} = 195$, $D_6 = \{5\}$ and $s_6 = 195 - \lfloor 4096/31 \rfloor = 63 \leq$

64. This is one row satisfying the bound below, not evidence that such rows occur at unbounded depths.

Write $r_n = r_n(1/21)$ for the real greedy remainder and $A_{1/21}$ for its support. The condition $\mathcal{F}_{21}$ means that there exist integers $n, R_0, K_0 \geq 0$ with all of the following properties:

1. $r_n > R_n$, the set of positive exponents outside $A_{1/21}$ is finite, and every exponent greater than n belongs to $A_{1/21}$;
2. for every $R \geq R_0$, the integer rule with target T_{2R} and weights $q_{2R}(2), \dots, q_{2R}(2R)$ selects exactly the same exponents as the real greedy rule on $\{2, \dots, 2R\}$;
3. for every $K \geq K_0$, the interval $(K, 2K]$ contains an exponent of $A_{1/21}$.

The eventual all-selected behaviour makes some clauses redundant. We retain them to specify both the real greedy sequence and the comparison with its finite integer versions.

Theorem D.1 (integer-quotient tests for $1/21$). *The following statements hold.*

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1. $1/21 \in A$ if and only if $\mathcal{F}_{21}$ does not hold.
2. If there is an unbounded sequence of ranks R with $s_R \leq 2^R$, then $1/21 \in A$.
3. On $\mathcal{F}_{21}$, eventually $s_R > 2^R$, the boundary rank $R + 1$ belongs to D_{R+1} , and, for all sufficiently large R ,

$$\begin{aligned} D_{R+1} &= D_R \cup \{R + 1\}, \\ s_{R+1} &= 4s_R + \left\lfloor \frac{4(2^{2R} \bmod 21)}{21} \right\rfloor - 2c_{D_R}(2R + 1) - c_{D_R}(2R + 2) \\ &\quad - (2^{R+1} + 1). \end{aligned}$$

Here $c_{D_R}(m) = \#\{d \in D_R : d \mid m\}$.

Proof. Write $x = 1/21$, $G = A_{1/21}$ and $\delta = x - X_G(2) \geq 0$. If $x \notin A$, then $\delta > 0$. The real remainder tends to δ , so every sufficiently small weight is selected and eventually $r_n > R_n$. For the integer comparisons, normalise by 4^R . As long as the selectors agree through rank $d-1$, the remainders differ by at most $d/4^R$. Each fixed comparison is strict, since equality would give a finite representation of x . Choose d_0 with $w_d < \delta/2$ for $d > d_0$. For large R the comparisons through d_0 agree and $2R/4^R < \delta/2$, so induction forces both rules to select all remaining ranks through $2R$. This proves $\mathcal{F}_{21}$. A represented target has $r_n \leq R_n$ at every rank, giving the converse in (1).

For every R , the rounding errors satisfy

$$\left| x - X_{D_R}(2) - \frac{s_R}{4^R} \right| \leq \frac{R + 1}{4^R}. \quad (13)$$

Hence an unbounded sequence with $s_R \leq 2^R$ gives finite subseries values tending to x . Closedness of A proves (2), and Section B excludes a finite representation.

On $\mathcal{F}_{21}$, the eventual agreement gives $D_R = G \cap \{2, \dots, R\}$ and $D_{R+1} = D_R \cup \{R + 1\}$. The rounding bound implies $s_R/4^R \rightarrow \delta > 0$. Finally, use $q_{M+2}(d) = 4q_M(d) + 2\mathbf{1}_{d \mid M+1} + \mathbf{1}_{d \mid M+2}$, together with the analogous target identity and $q_{2R+2}(R + 1) = 2^{R+1} + 1$, to

obtain (3). The full comparison induction and both quotient identities are given in [the long record's integer-quotient proof for 1/21](#). $\square$

The finite uniqueness statement is also explicit: if $D \subseteq \{2, \dots, R\}$ and an integer s satisfy $\sum_{d \in D} q_{2R}(d) + s = T_{2R}$ with $0 \leq s \leq 2^R$, then $D = D_R$ and $s = s_R$. This is the denominator-specific separation theorem ([uniqueness of a finite representation with the stated remainder bound](#)). The cited finite crossing lemmas give additional consequences under their alignment hypotheses: an earlier finite prefix cannot occur, and a real greedy exponent must be skipped ([the missing-prefix consequence of an aligned crossing](#), [the real greedy skip forced by that crossing](#)).

Approximation to 1/2 by finite supports. For $A \subseteq \{2, \dots, M\}$ and $1 \leq m \leq M$, define the integer

$$K_A(m) = 2^{m-1} - \sum_{j=2}^m 2^{m-j} c_A(j).$$

It measures the error in the truncated divisor-coefficient sum, after multiplication by 2^m . In particular, $K_A(1) = 1$ and $K_A(m+1) = 2K_A(m) - c_A(m+1)$.

Theorem D.2 (finite approximations with vanishing scaled error). *Suppose there are integers $M_j \geq 1$ tending to infinity and sets $A_j \subseteq \{2, \dots, M_j\}$ such that* [Lean](#)

$$\frac{|K_{A_j}(M_j)|}{2^{M_j}} \rightarrow 0.$$

Then $X_A(2) = 1/2$ for some infinite set $A \subseteq \mathbb{N}_{>0}$.

To compare the finite subseries with 1/2, we separate the coefficient tail from the terminal carry. No agreement between the supports A_j or bound on their earlier carries is needed. The estimate is

$$\left| X_{A_j}(2) - \frac{1}{2} \right| \leq \frac{|K_{A_j}(M_j)| + 2\sqrt{M_j} + 4}{2^{M_j}}.$$

Indeed, $c_{A_j}(n) \leq \tau(n) \leq 2\sqrt{n}$ and $\sqrt{M+r} \leq \sqrt{M} + r$, so the normalised tail is at most $\sum_{r \geq 1} 2^{-r} (2\sqrt{M} + 2r) = 2\sqrt{M} + 4$. The finite sums therefore tend to 1/2. The achievement set is closed, and no finite support has value 1/2, so its representing support is infinite. A square-root bound on the terminal carry would suffice, but the theorem permits any error $o(2^{M_j})$. For a finite $D \subseteq \{2, \dots, M\}$, the floor-quotient identity gives

$$K_D(M) = 2^{M-1} - \sum_{d \in D} \left\lfloor \frac{2^M}{2^d - 1} \right\rfloor.$$

The exact quotient rows considered in the long record satisfy $\sum_{d \in D} q_M(d) = 2^{M-1} - 1$, so they have $K_D(M) = 1$. The terminal criterion allows a larger error of either sign. At a fixed depth these are different conditions, although cofinal existence of either kind of approximation characterises half-membership.

Conversely, if $X_A(2) = 1/2$, then $1 \notin A$, and the prefixes $A_M = A \cap \{2, \dots, M\}$ satisfy

$$K_{A_M}(M) = \sum_{r \geq 1} c_A(M+r)2^{-r} \leq 2\sqrt{M} + 4.$$

The equality uses the coefficients of the full support A on the right: truncation does not change the coefficients through M . Hence the existence hypothesis of Theorem D.2 is equivalent to half-membership. No compatible sequence of prefixes is required for the forward implication. The existence of the approximating supports remains unproved and would refute Problem 1.1.

Finite families with a shared prefix. Put $B(m) = 2\lfloor \sqrt{m} \rfloor + 4$. For $0 \leq K \leq M$, consider a family $A_1, \dots, A_{B(M)} \subseteq \{2, \dots, M\}$ with

$$1 \leq K_{A_k}(m) \leq B(m) \quad (1 \leq m \leq M), \quad K_{A_k}(M) = k.$$

Require that all the A_k agree on $\{1, \dots, K\}$ and that some integer $E \geq B(M)$ satisfies

$$\sum_{j=K+1}^M 1_{A_k}(j) 2^{M-j} + k = E \quad (1 \leq k \leq B(M)).$$

Thus the binary suffixes have consecutive values $E - k$, while the terminal carry ranges through the entire permitted interval. This is substantially more data than one finite approximation to $1/2$.

Theorem D.3 (unbounded shared-prefix families imply a half-support). *Suppose that for every N there are M, K with $\max\{N, 1\} \leq M$, $0 \leq K \leq M$, and a family satisfying all the conditions in the preceding paragraph. Then $X_A(2) = 1/2$ for some infinite set $A \subseteq \mathbb{N}_{>0}$.*

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For each M choose the member with terminal carry 1. Its error 2^{-M} tends to zero along unbounded depths, so Theorem D.2 applies. The other shared-prefix conditions are additional requirements of the proposed construction; the compactness argument uses only this one member. No construction at unbounded depths is established here.

The tests above leave three kinds of arithmetic information to be supplied: returns of the actual greedy sequence, finite approximants at unbounded depths, or an exclusion of a possible final skip. The following questions specify those missing inputs.

Problem D.4 (membership of $1/21$ in the Mersenne achievement set). For the greedy remainder of $1/21$, prove that arbitrarily large N satisfy

$$2^{N+1}r_N(1/21) < \frac{2^{N+1}}{2^{N+1}-1}, \quad \text{equivalently} \quad r_N(1/21) < \frac{1}{2^{N+1}-1}.$$

Equivalently, exclude the condition $\mathcal{F}_{21}$ defined above. One sufficient route is to rule out the eventual recurrence in Theorem D.1(3) together with $s_R > 2^R$; another is to prove $s_R \leq 2^R$ at arbitrarily large ranks. Neither sufficient route is claimed to be necessary by itself.

Problem D.5 (bounded returns of the scaled remainder). For each of the targets $x = 1/2$ and $x = 1/21$, does the greedy remainder $r_N(x)$ return to one bounded interval after scaling by 2^N ?

$$\exists B < \infty \forall K \exists N \geq K : \quad 2^N r_N(x) \leq B.$$

Problem D.6 (arithmetic tests for the greedy sequence). Can a finite-memory, 2-adic or discrepancy argument prove $s_R \leq 2^R$ at arbitrarily large ranks, or rule out the eventual recurrence in Theorem D.1(3)? Such an argument must use the divisor counts of the actual greedy support. Can a bounded window of $R \bmod 6$, residues of s_R , endpoint divisor counts and the finite set of eventual skips force a decrease or a contradiction? Alternatively, can one show that these bounded-memory data cannot distinguish the actual sequence from sequences that remain above 2^R but need not come from a support?

Problem D.7 (final-skip Diophantine exclusion). Let $E = \sum_{n \geq 1} (2^n - 1)^{-1}$. If $1/21 \notin A$, let M be the last skipped exponent, S_M its finite skipped prefix, and

$$a_M = \frac{1}{21} + \sum_{d \in S_M} \frac{1}{2^d - 1}.$$

Write $\text{gap}_M = (2^M - 1)^{-1} - R_M > 0$. Can the arithmetic restrictions on this last skipped exponent be used to prove $|E - a_M| \geq \text{gap}_M$, contradicting

$$0 < a_M - E < \text{gap}_M?$$

E Elementary details used in the proofs

Cyclotomic prime-power fact. Let $b \geq 2$, $n \geq 2$, $\ell \mid \Phi_n(b)$ be prime and $e = v_\ell(b^n - 1)$. Then $\text{ord}_{\ell^e}(b) = n$. We include the 2-adic case. For odd ℓ , put $d = \text{ord}_\ell(b)$ and $s = v_\ell(b^d - 1)$. The elementary lifting identity $v_\ell(b^{dt} - 1) = s + v_\ell(t)$ follows by factoring a geometric sum when $\ell \nmid t$ and by a binomial expansion for a factor ℓ . In the factorisation $b^n - 1 = \prod_{r \mid n} \Phi_r(b)$, subtracting these valuations over proper divisors gives

$$v_\ell(\Phi_n(b)) = \begin{cases} s & n = d, \\ 1 & n = d\ell^j, j \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

If $n = d$, no smaller positive exponent gives $b^m \equiv 1 \pmod{\ell}$. If $n = d\ell^j$, the full valuation is $e = s + j$; the same lifting identity shows that the least exponent giving valuation e is $d\ell^j = n$. For $\ell = 2$, b is odd. Factoring $b^{2^t} - 1$ gives

$$v_2(b^{2^j} - 1) = v_2(b - 1) + v_2(b + 1) + j - 1 \quad (j \geq 1).$$

The only possible indices $n \geq 2$ are $n = 2^j$: $v_2(\Phi_2(b)) = v_2(b + 1)$ and $v_2(\Phi_{2^j}(b)) = 1$ for $j \geq 2$; all other indices have valuation zero. The displayed identity again makes 2^j the least exponent with the full valuation. This proves the fact. Finally $\Phi_n(b) > 1$: each primitive n th root ζ satisfies $|b - \zeta| > 1$, and their product is $\Phi_n(b)$.

Exact scalar cover cost. For $t \geq 1$, put $z = \log_2 t$. The substitution $y = 2^\alpha$ reduces the infimum defining $\Psi(t)$ to $y^z/(y-1)$ on $1 < y \leq 2$. Its derivative has the sign of $(z-1)y - z$. Hence

$$\Psi(t) = \begin{cases} t, & 1 \leq t \leq 4, \\ z^z/(z-1)^{z-1}, & t > 4. \end{cases}$$

The endpoint and the open lower bound $\alpha > 0$ are both included in this calculation. This scalar identity does not supply a converse to the positive-cover criterion.

F Sources

Formal proofs. A result with a kernel-checked Lean proof carries a mark in the margin. *Lean* opens the proof: the declaration itself when one declaration states the whole result, otherwise the list of declarations that together state it. *Comparator* opens the record of an independent check, in which the same statement, written again from Mathlib alone in a separate repository, was compared with our proof by Lean’s Comparator tool, allowing only the three standard axioms. A dagger on the Lean mark means that the Lean proof assumes an input named just below the result. A result without a mark has no Lean proof of its whole statement; what is checked is said below it. The [evidence record](#) gives every declaration, version and check. These checks show that the stated propositions are proved; whether they are the right propositions is a question the reader can settle by comparing them with the text.

For Erdős’s reciprocal-summable extension and fractional-part argument, see pp. 222 and 226 of [7]. The periodic theorem [12] is also stated in Luca and Tachiya’s account [11, Theorem A and Example 2, pp. 139–140], and Hornich’s strict-tail theorem is proved in Nitecki’s exposition [10, Theorem 4(1)]. These expositions are the sources used here for those two results; the original articles were not independently retrieved. The *Formal Conjectures* file [6] is statement-level prior art, not a proof dependency.

The linked evidence record identifies the exact Lean statements, source revisions and recorded Comparator checks. No such checks were rerun for this revision. The ordinary arguments here have not had independent human review. They include the $A_\star$ calculation, the fresh-prime construction, the signed finite-denominator extension and the proof of Theorem D.1. For the incomparability result, the formal construction of V uses a different squarefree divisor-cube schedule. The mixed implication is also formalised. The three logarithmic-sampling results in the long record have separate comparison records; these do not constitute human review of the ordinary proofs. No historical priority is claimed for the support comparison. Conditional statements retain their antecedents, and computations retain their finite ranges.

Reproducing the weighted theorem. The two declarations for Theorem 1.2 are in the public source at [commit 91ca3405](#). That snapshot pins Lean 4.29.1 in `lean-toolchain` and its Mathlib revision in `lake-manifest.json`. From a complete checkout, with `elan` installed, build the two modules with:

```
lake exe cache get
python3 scripts/lean_fast_build.py --jobs 2 \
```

```

ErdosProblems.Erdos257.PaperCompleteR8.WeightedReturn \
ErdosProblems.Erdos257.PaperCompleteR8.WeightedHereditaryClaim

```

The cache download is optional. The modules contain `divisibilityWeightedClaim` and `finitePrimeWeighted_fixedBase_hereditary`, respectively; their precise statements and associated checks are linked above.

The supplementary declarations are collected in [the long record's final source section](#), grouped by support criteria, finite denominators, integer recurrences and achievement sets.

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